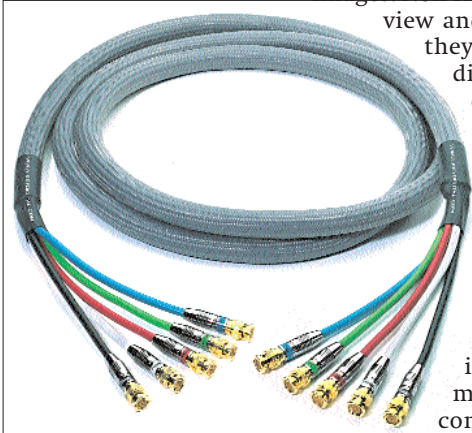


Now that we're clear on the basics of TV and HDTV, let's move on to the specifics of what makes a High-Definition television.

One Big Family

The majority of HDTV sets, be they LCD, plasma, rear-projection microdisplays, DLP, or LCoS technology, are so-called fixed-pixel displays. Such a display uses pixels to form images. Non-fixed-pixel displays are direct-



HDMI to DVI-D adapter

view and rear-projection CRTs, and they don't use discrete pixels to display images. Fixed-pixel HD displays operate at a "native" resolution, say 720p. They are also all inherently progressive-scan—even a 1080i "interlaced" HDTV set runs at a 720p "progressive" resolution. Since an HDTV only talks in native resolution, there is some scaling involved with content: no matter what the source of the content, a fixed-pixel HDTV will always scale it to fit to its native resolution.

If the incoming signal has more pixels than the TV can handle, you will lose some detail and sharpness. If in turn the signal has fewer pixels, the TV will scale it to fit its screen but you won't get any extra detail and sharpness (this is called upscaling or upconverting). How well the TV does at scaling is a critical point to consider while purchasing an HDTV.

Before making a purchase, we recommend that you play various content on your desired set—TV, DVD, HD video games, and if available, HD movies.

What Flavour HDTV?

Wouldn't it be great if you could just walk into a store and pick up an HDTV, any HDTV, without worrying about black levels, contrast ratio, refresh rates, viewing angles, colour saturation, colour accuracy and so on... of course it would, but if you're in the market for an HDTV, you need to understand the pros and cons of each HDTV technology.

Broadly speaking, there are two major categories of HDTV—you have your Flat Panel TV sets, which are your thin plasmas and LCD panels. Then you have your rear-projection sets which can be LCD, DLP (Digital light Processing), CRT, or LCoS (Liquid crystal on silicon). Apart from these, you can also purchase an HD-resolution projector, or an old-fashioned CRT TV set. Let us run through the major technologies first, and then take a look at the table *Which HDTV Should I Buy?*

Flat Panel Displays

There is little to differentiate between a plasma and an LCD display today—each one's pro is generally replicated well by the other. A plasma screen is no longer as liable to suffer from burn-in as it used to, while an LCD TV's viewing angle is much improved today than it was a year ago. And so on. Generally speaking, a plasma television is still a better choice if you are looking to set up a home-theatre: these are large screens with great image quality, high brightness, and good contrast ratio. The only negative is screen brightness: a plasma TV will gradually dim over its life. Present plasmas are typically 60,000 hrs of half-life—long enough to last five to six years, but something to keep in mind while picking one up. For gaming, both technologies have their cons: an LCD screen is liable to "ghost" or blur when the action gets fast (due to slower response time); a game's HUD (Head-up Display) is liable to

Which HDTV Should I Buy?

HDTV technology		What's great about it	What sucks
Flat Panel TV	Tube televisions (non-fixed-pixel displays)	Inexpensive, wide viewing angle, great picture quality (especially black levels)	Huge, extremely heavy, limited screen size, lower resolution
	Plasma television	Thin, wide viewing angle, image quality ideal for a home theatre setup, faster response time than LCD, brighter than LCD	Slight potential for burn-in, more expensive than LCD, display gradually dims over time (loses brightness)
	LCD television	Widest range of sizes available, no danger of burn-in, higher resolution than comparably-sized plasmas, generally cheaper than plasma	Image quality generally not as good as plasma (poor blacks); relatively narrow viewing angle, might show ghosting while displaying fast-moving images like in games
Rear Projection TV	CRT	Inexpensive, good black levels, good colour saturation, sharp picture	Need professional calibration for ideal setup, deep cabinets, not suited for bright rooms
	Microdisplays	DLP (Digital Light Processing)	Rainbow effect might be visible in some sets, low-level noise, wobulated resolution on some TVs
		LCD	1080p currently unavailable, black level lacking, uniformity issues, not as bright as DLP, currently no 1080p chips, poor contrast ratios, weak and stuck pixels are common
		LCoS	Some uniformity issues
Projectors (Front Projection)	DLP	Better black level than LCD, good uniformity, good brightness	Rainbow effect, more expensive than LCD
	LCD	Usually cheaper and more feature-packed than DLP of same resolution	Not as bright as DLP, might show screen-door effect